

Agent Windows Service Installation Guide

Step-by-step setup for each company QuickBooks machine

This guide installs the M2B sync agent as a Windows Service so it starts with Windows, reads balances from QuickBooks Desktop (or mock mode), and pushes snapshots to your Laravel API. The mobile app never talks to QuickBooks - only the agent does.

Before you start

You need: Administrator PowerShell, .NET 8 runtime, backend reachable (e.g. <http://127.0.0.1:8000>), Filament admin login, and (for live mode) QuickBooks Desktop Enterprise.

What you will do

- 0 Start backend API
- 1 Create agent token in Filament
- 2 Install Windows Service (Admin)
- 3 Write Production config + token
- 4 Authorize QuickBooks (live only)
- 5 Restart service & verify logs
- 6 Optional: test mobile app

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Start the backend API

Leave this terminal running

In a normal (non-admin) PowerShell window:

```
cd "C:\Users\lappybooks\MurabacApps\M2B Ledger\backend"  
php artisan serve --host=127.0.0.1 --port=8000
```

- > Open <http://127.0.0.1:8000/up> - you should get a healthy response.
- > Admin panel will be at <http://127.0.0.1:8000/admin>

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Create an agent token in Filament

Shown once - copy it immediately

- > Browse to <http://127.0.0.1:8000/admin>
- > Login: `admin@demo.test` / `Password123!`
- > Left nav -> Agent tokens -> Create / New
- > Name it e.g. Local PC or Company Name
- > After save, COPY the plaintext token once (it is never shown again)
- > Demo Company id is usually 1

Security

Never commit the token to git. Put it only in `appsettings.Production.json` on the machine (that file is gitignored) or in environment variables `Agent__AgentToken`.

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Install the Windows Service

Must use Administrator PowerShell

Common mistake

If you see `"#requires Administrator"`, you are not elevated. If you see `"module agent could not be loaded"`, you are already inside the agent folder - use `.\scripts\install-service.ps1` instead of `agent\scripts\...`

1. Start menu -> Windows PowerShell -> right-click -> Run as administrator
2. From the repo root:

```
cd "C:\Users\lappybooks\MurabacApps\M2B Ledger"  
.\agent\scripts\install-service.ps1
```

Or if your prompt is already inside the agent folder:

```
.\scripts\install-service.ps1
```

If PowerShell blocks scripts:

```
Set-ExecutionPolicy -Scope Process Bypass  
.\agent\scripts\install-service.ps1
```

What the script does

- > Publishes the agent as win-x86 to C:\Program Files\QbBalancesAgent
- > Creates Windows Service QbBalancesAgent (display: QB Balances Sync Agent)
- > Sets StartupType = Automatic
- > Does NOT start the service yet - configure the token first

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Write Production config + token

Do NOT edit the repo appsettings.json for the service

The service reads config from the install folder. Create Production settings there:

```
notepad "C:\Program Files\QbBalancesAgent\appsettings.Production.json"
```

Paste this template and replace PASTE_TOKEN_HERE with your Filament token:

```
{
  "Agent": {
    "BackendUrl": "http://127.0.0.1:8000",
    "AgentToken": "PASTE_TOKEN_HERE",
    "CompanyFilePath": "",
    "PollIntervalSeconds": 180,
    "MockMode": true,
    "CompanyId": 1,
    "AppName": "QB Balances Sync Agent",
    "CustomerMaxReturned": 100
  }
}
```

Key settings

Setting	Meaning
MockMode: true	Fake data - good first test (no QB needed)
MockMode: false	Live QuickBooks COM read
CompanyFilePath: ""	Use the company file currently open in QB
CompanyId: 1	Must match Filament company id
BackendUrl	Use https://... in real production hosting

Tip

Prove the pipeline with MockMode true first. When Filament Sync logs show success, flip MockMode to false, open QuickBooks, restart the service, and authorize the app.

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Authorize QuickBooks (live mode only)

Skip while MockMode is true

- > Open QuickBooks Desktop Enterprise and open the company file
- > Start or restart the service so QB can prompt

```
Start-Service QbBalancesAgent  
# or  
Restart-Service QbBalancesAgent
```

- > When prompted for QB Balances Sync Agent, choose always allow / unattended access
- > Prefer a view-only QuickBooks user
- > Confirm: Edit -> Preferences -> Integrated Applications -> Allowed

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Restart service & verify

Silent success is normal - check logs

Why Start-Service looks like "nothing happened"

Windows Services do not print to your PowerShell window. Get-Service and the log file are how you verify.

```
Restart-Service QbBalancesAgent
Get-Service QbBalancesAgent
Get-Content "C:\Windows\SysWOW64\logs\agent-*.log" -Tail 40
```

Because the agent is published as win-x86, file logs often land under C:\Windows\SysWOW64\logs\ (not Program Files). Look for HTTP 200 / Snapshot accepted.

- > Also check Filament -> Sync logs -> latest row = success
- > 401 Unauthorized = missing/wrong AgentToken - fix Production json and restart
- > Empty AgentToken warning in log = Step 3 not done

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Optional: test the mobile app

Refresh only reloads the API snapshot

```
adb reverse tcp:8000 tcp:8000
cd "C:\Users\lappybooks\MurabacApps\M2B Ledger\mobile"
flutter run --dart-define=API_BASE_URL=http://127.0.0.1:8000
```

- > Login example: owner@demo.test / Password123!
- > Pull to refresh after a successful agent cycle
- > Phone refresh does NOT sync QuickBooks - only the agent does

Useful commands

Action	Command
Status	Get-Service QbBalancesAgent
Start	Start-Service QbBalancesAgent
Stop	Stop-Service QbBalancesAgent
Restart	Restart-Service QbBalancesAgent
Uninstall	.\agent\scripts\uninstall-service.ps1
Tail logs	Get-Content ...\SysWOW64\logs\agent-*.log -Tail 40

Architecture reminder

Company QB PC -> Agent (this service) -> HTTPS POST /api/agent/sync -> Laravel API -> one Play Store app. Each company gets its own agent token and users (company_id).

More docs

README.md - docs/TESTING.md - agent/README.md - docs/ARCHITECTURE.md